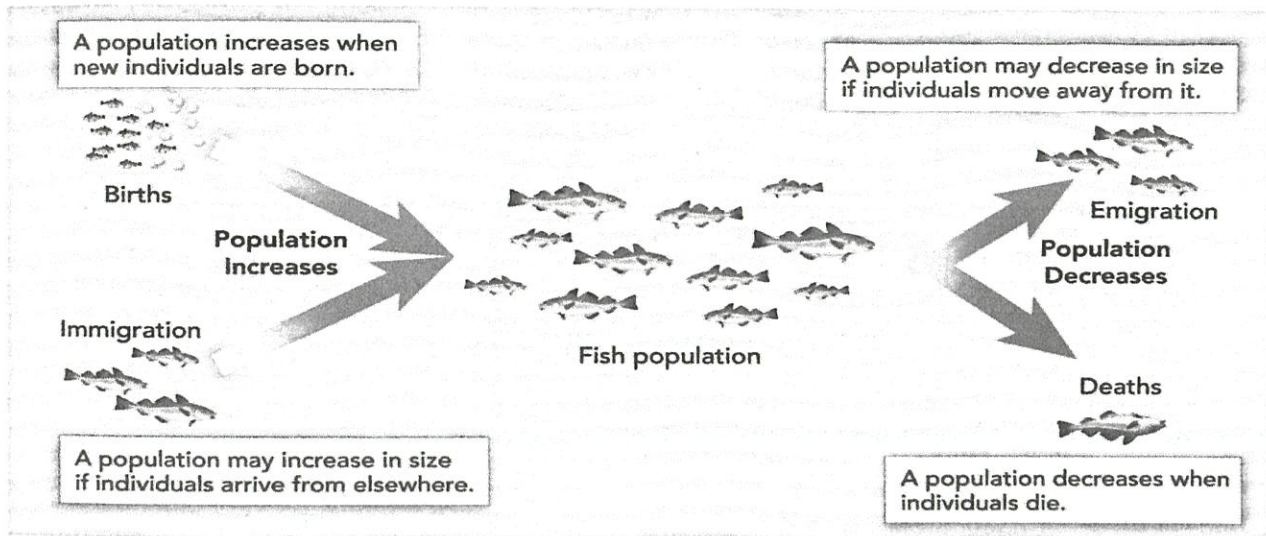


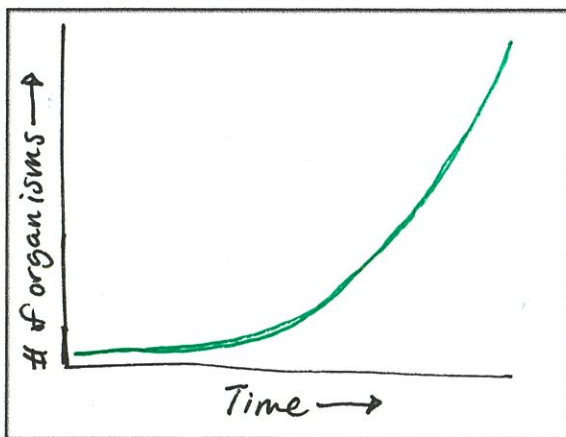


Use the figure 2-3 above (also found on pg.41) to answer the questions below:

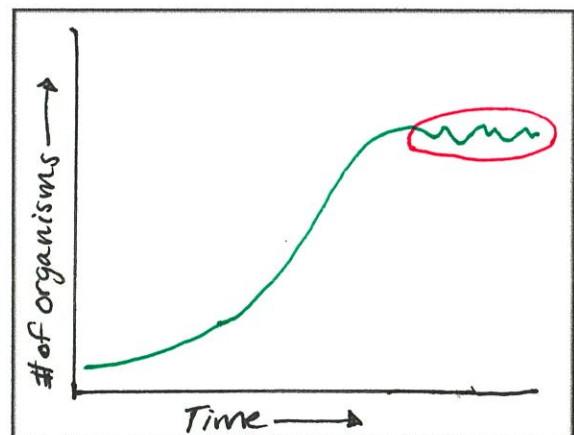
5. What two factors add individuals to the fish population? Births & Immigration
6. What two factors remove individuals from the fish population? Deaths & Emigration
7. If the fish population stays the same size for a one-year period, what can you assume about the number of individuals removed from the population because of death and emigration during that time? That number is equal to the number of individuals added to a pop. by birth or immigration
8. Assuming this fish example was demonstrating an ocean species, what factor(s) would change about the population if these fish were found in a large pond? immigration / emigration would not factor into a populations growth rate as there is no place to swim to or from (closed system)

Exponential and Logistic Growth

9. When does exponential growth occur? under ideal conditions with unlimited resources.
→ each generation produced contains more individuals than previous one.
10. What are 'ideal conditions' for a population to grow exponentially? unlimited food & space.
protection from disease / predators. Removal of waste products.
11. Use figures 2-4 and 2-5 to draw both a J-shaped curve and a S-shaped curve. CIRCLE where the carrying capacity is. Include axis labels as well



Exponential Growth



Logistic Growth

12. Cause and Effect

Write the *effect* for both types of growth patterns. Include terms like; birth & death rates.

Type of Population Growth		Cause	Effect
Exponential Growth		A species is introduced into a new region with plentiful resources and few predators.	Most species will survive & produce offspring. Each generation will produce more than previous. = RAPID GROWTH
Logistic Growth	Phase 1	A species is introduced into a new region with plentiful resources and few predators.	Population grows rapidly in most cases. Exponentially.
	Phase 2	Resources become limited.	Population still grows but starts to slow as resources are harder to come by.
	Phase 3	The carrying capacity of the environment is reached.	rate of population growth stops. There may be slight increase/decrease but overall similar rate.

13. Explain **carrying capacity** for a population: The number of individuals of a species that an environment can support of that species.
14. Looking back at figure 2-5 (or even your own graph), how is the carrying capacity of a population shown? with a dashed line that is the average # of individuals once population is stabilized.
15. List 2 specific examples of biotic and abiotic environmental factors that affect carrying capacities:
- **Biotic:** amount of space can limit the amount of barnacles or mussels that can attach to rocks.
 - **Abiotic:** amount of water available could determine # of species, even size of species.